

Natural Language Processing to Extract Antimicrobial Resistance Patterns From Unstructured, Synthetic Microbiology Reports

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Abstract

Antimicrobial resistance is a global health concern and unstructured laboratory reports contain valuable data for monitoring trends.

Synthetic data were generated using large language model ChatGPT and the microbiology_cultures_labs.csv (January 2025) from the Antibiotic Resistance Microbiology Dataset. This is de-identified data taken from electronic health records at Stanford Healthcare.

Natural Language Processing (NLP) models were used to extract organism, antibiotic, susceptibility and resistance groups, focusing on Methicillin-resistant *Staphylococcus aureus*, Carbapenem-resistant Enterobacterales, and Vancomycin-resistant Enterococcus.

Three approaches were evaluated: classification of resistance groups from directly from text, rule-based classification from extracted variables, and multi-output models including Logistic Regression, Support Vector Machines (SVM) and Naive Bayes.

Text Cleaning was compared against raw inputs and improved performance. The SVM model achieved $F1 > 0.95$ across all tasks, demonstrating that NLP can effectively extract information from unstructured clinical text.

1 Introduction

Antimicrobial resistance (AMR) is a global health concern recognised by the World Health Organization and contributed to 4.95 million deaths in 2019 (World Health Organization, 2023). The rise of drug-resistant pathogens threatens the treatment of common infections and the achievements of modern medicine, making procedures such as surgery increasingly unsafe.

Monitoring AMR helps prevent the spread of resistant strains in hospitals, supports research, and preserves the future effectiveness of antibiotics. This relies on microbiology laboratory data, where antibiotic susceptibility testing is performed

on identified organisms. However, reports are often unstructured, making automated extraction a challenge due to abbreviations and inconsistent formatting.

NLP enables the extraction of structured information, including bacteria and antibiotic susceptibility data. NLP supports large-scale analysis and integration into standardised shared databases such as the Observational Medical Outcomes Partnership (OMOP) (White et al., 2019), where the shared information can be used for research and prevention purposes.

1.1 Related work

Synthetic data generated by large language models (LLMs) has emerged as solution to the limited access of health data. However, studies raise concerns about the realism, bias, and reproducibility of synthetic outputs (Chen et al., 2021).

Clinical NLP has traditionally relied on rule-based systems to extract structured information from free-text reports, particularly in microbiology because they reflect clinical decision logic. Machine learning approaches using representations such as term frequency-inverse document frequency (TF-IDF) have improved classification performance, and preprocessing and text normalisation has shown to significantly impact results (Li et al., 2021).

More recent techniques involve integrated modeling approaches, including multi-task learning and LLM-based systems, which predict multiple outputs simultaneously and capture relationships within high-dimensional clinical data (Ruder, 2017). This is relevant in AMR classification, where labels depend on combinations of organism, antibiotic, and susceptibility. While rule-based systems explicitly encode these relationships, joint learning approaches may improve consistency and capture underlying dependencies within complex datasets.

This study applies NLP approaches to synthetic data created by LLM ChatGPT to extract organism, antibiotic, and susceptibility information and classify resistance groups, with a focus on Methicillin-resistant *Staphylococcus aureus* (MRSA), Carbapenem-resistant Enterobacterales (CRE), and Vancomycin-resistant Enterococcus (VRE).

1.2 Ethical considerations

Clinical data in the UK is protected under the UK General Data Protection Regulation (GDPR, 2018) and governed by organisations such as the National Health Service (NHS), who restrict access to identifiable patient information.

Due to these constraints, the Antibiotic Resistance Microbiology Dataset (ARMD) (Nateghi Haredasht et al., 2025) was used to generate synthetic data. ARMD is a publicly available, de-identified dataset derived from electronic health records at Stanford Healthcare. The dataset was developed under approval from the Stanford University Institutional Review Board (eProtocol #70466), which classified the study as minimal risk and waived patient consent due to the use of fully de-identified retrospective data.

2 Data

This project used the microbiology_cultures_cohort.csv file (January 2025) (Nateghi Haredasht et al., 2025) from the ARMD, which contained microbiology culture records from over 283,000 adult patients. Variables of interest included organism, antibiotic, and susceptibility.

An exploratory data analysis was performed to examine the distribution of organisms. The resistance group labels were not present, so the data were filtered to construct three clinically defined datasets: MRSA, VRE, and CRE. MRSA cases were defined as *Staphylococcus aureus* resistant to oxacillin (n = 936), CRE as *Escherichia coli* and *Klebsiella pneumoniae* resistant to meropenem, imipenem, or ertapenem (n = 102), and VRE as *Enterococcus faecium* and *Enterococcus faecalis* resistant to vancomycin (n = 179). A label column was added based on these rules.

Each dataset was balanced with equal numbers of positive and negative samples, with 500 samples per class for MRSA, 100 per class for CRE and 100 per class for VRE. The negative class was

subdivided into five groups (20% each): (A) target organisms sensitive to relevant antibiotics, (B) other organisms sensitive or (C) resistant, to the relevant antibiotic, (D) similar organisms e.g. coagulase negative *Staphylococcus* (sensitive or resistant), and (E) random sensitive organisms. This approach prevented the model from relying on organism identity alone, and encouraged models to learn clinically meaningful resistance patterns.

The filtered datasets were uploaded to ChatGPT Auto (versions 5.3/5.4 on 15/03/2026 at 17.00 (OpenAI, 2026) with the prompt "Create synthetic clinical style report data that MUST include organism, antibiotic and sensitivity". However, the text was still fairly structured. Further prompts were added including "include specimen type, site of infection, abbreviations, variations, company names, official names, outdated names, consider non-native writers and clinical sort-hand". Data were manually reviewed against requirements and no hallucinations were found.

3 Methods

All data processing, analysis, and modelling were carried out in Python (version 3.14), using pandas and NumPy for data handling and the scikit-learn library (version 1.8.0) for machine learning and all random states were fixed at 42.

A cleaned text column was created to compare the effect of preprocessing on model performance. Text was converted to lowercase and punctuation was removed except for those used in clinical practice "/", ".", and "-". Regex-based normalisation and standardisation rules were applied to popular short-hand terms (e.g. *E. coli* to *Escherichia coli*).

Data were split into training and testing sets using an 80:20 stratified split. Text was vectorised using TF-IDF with unigrams and bigrams (ngram_range = (1,2)) and models were trained with a maximum of 2000 iterations.

Three approaches were evaluated. In the direct approach, logistic regression models were trained to classify resistance groups directly from raw and cleaned text. Both binary (MRSA vs NOT_MRSA, VRE vs NOT_VRE, CRE vs NOT_CRE) and multiclass (MRSA, VRE, CRE, NONE) models were developed.

In the extraction approach, separate logistic regression models were trained to predict organism, antibiotic, and susceptibility from raw and clean text. A rule-based model was then applied to clas-

sify resistance groups from extracted variables. The rules were based on the organism–antibiotic susceptibility definitions defined previously that were used to construct the datasets.

In the joint approach, Logistic Regression, Support Vector Machine (LinearSVC), and Naive Bayes models were developed using a MultiOutputClassifier to simultaneously predict organism, antibiotic, susceptibility, and resistance group from the same input, which was the clean text.

All models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices to review misclassified examples.

4 Results

Direct classification results are shown in Figure 1 (left) and Table 1. All models achieved high performance ($F1 > 0.94$), with all models trained on cleaned text outperforming those trained on raw text. For raw text, the combined multiclass model performed best, whereas for cleaned text the CRE binary model achieved the highest performance.

Dataset	Text	Acc	Prec	Rec	F1
CRE	Clean	0.975	0.976	0.975	0.975
	Raw	0.950	0.955	0.950	0.950
Combined	Clean	0.971	0.972	0.971	0.971
	Raw	0.957	0.958	0.957	0.956
MRSA	Clean	0.940	0.942	0.940	0.940
	Raw	0.930	0.931	0.930	0.930
VRE	Clean	0.950	0.955	0.950	0.950
	Raw	0.925	0.926	0.925	0.925

Table 1: Performance of models that classified resistance group directly from clean and raw synthetic microbiology report text.

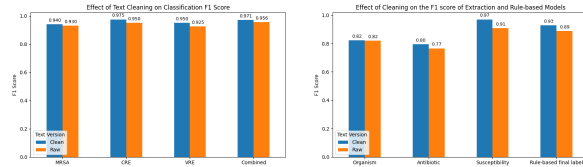


Figure 1: The effect of cleaning on the F1 scores for models that classified resistance group: directly from synthetic microbiology report text (Left) and using a rule based model from extracted organism, antibiotic and sensitivity variables (Right)

Results for the extraction and rule-based approaches are illustrated in Figure 1 (right) and Table 2. Models trained on cleaned text again outperformed those trained on raw text across all tasks. Susceptibility classification achieved the highest performance ($F1 = 0.970$), while rule-based classification of resistance groups also performed well ($F1 = 0.928$). However, organism and antibiotic extraction showed lower performance, with F1 scores of 0.822 and 0.795, respectively.

Task	Text	Acc	Prec	Rec	F1
Organism	Clean	0.861	0.816	0.861	0.822
	Raw	0.861	0.814	0.861	0.821
Antibiotic	Clean	0.843	0.787	0.843	0.795
	Raw	0.821	0.759	0.821	0.766
Susceptibility	Clean	0.975	0.972	0.975	0.970
	Raw	0.914	0.913	0.914	0.908
Rule-based	Clean	0.929	0.938	0.929	0.928
	Raw	0.889	0.907	0.889	0.888

Table 2: Performance of models that predicted organism, antibiotic, susceptibility from raw and clean text, and the rule-based model was applied to classify resistance group.

The performance of the joint models is shown in Figure 2 (left) and Table 3. The SVM model outperformed Logistic Regression and Naive Bayes across all tasks, and Naive Bayes consistently showed the lowest performance. The SVM scored $F1 = 0.986$ for resistance group classification and performed strongly across organism, antibiotic, and susceptibility prediction ($F1 > 0.95$). The confusion matrix in Figure 2 (right) shows minimal misclassifications for the SVM model, with only a small number of false positives observed.

Task	Model	Acc	Prec	Rec	F1
Organism	LogReg	0.861	0.816	0.861	0.822
	SVM	0.961	0.943	0.961	0.950
	NB	0.779	0.640	0.779	0.697
Antibiotic	LogReg	0.843	0.787	0.843	0.795
	SVM	0.993	0.989	0.993	0.991
	NB	0.729	0.604	0.729	0.642
Susceptibility	LogReg	0.975	0.972	0.975	0.970
	SVM	0.993	0.993	0.993	0.993
	NB	0.857	0.852	0.857	0.843
Resistance Group	LogReg	0.971	0.972	0.971	0.971
	SVM	0.986	0.986	0.986	0.986
	NB	0.900	0.906	0.900	0.900

Table 3: Performance of integrated models that simultaneously predicted organism, antibiotic, sensitivity and resistance group from cleaned synthetic microbiology reports.

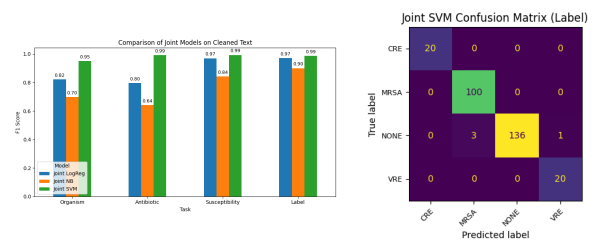


Figure 2: Comparison of integrated models that simultaneously predicted organism, antibiotic, sensitivity and resistance group from cleaned synthetic microbiology reports (Left) and SVM classification performance for resistance groups (Right)

5 Discussion

Direct classification tasks performed well in all cases ($F1 > 0.94$), however this did not extract any useful information about the organism and antibiotic. The combined model ($F1 = 0.971$) would be

recommended compared to multiple binary models ($F1 = 0.940\text{--}0.975$) as it is more simple to use and maintains performance.

Extraction and rule based approaches performed well, but performance was again lower for antibiotic ($F1 = 0.795$) and organism ($F1 = 0.822$). Naive Bayes models consistently performed lower compared to other approaches ($F1$ as low as 0.697). This may be because Naive Bayes models assume feature independence and the resistance class is dependent on antibiotic, organism and susceptibility (McCallum and Nigam, 1998). Whereas SVM was able to learn complex decision boundaries from high-dimensional text where studies have shown SVM's perform strongly in high-dimensional clinical text classification tasks (Cortes and Vapnik, 1995).

Susceptibility prediction performed well across all models ($F1 = 0.970\text{--}0.993$), which is likely due to the smaller number of classes (Resistant, Sensitive, Intermediate), whereas antibiotic and organisms were much more complex. High model performance could be due to synthetic data being more structured and less noisy than real-world clinical text. Studies show reduced variability can lead to inflated model performance (Chen et al., 2021).

Figure 1 shows text cleaning improved performance across all direct and extract approaches, so was used for the joint model comparisons. This aligns with studies showing that preprocessing and normalisation significantly improve performance in clinical NLP tasks (Li et al., 2021).

The SVM model achieved the highest performance ($F1$, accuracy, recall and precision = 0.986 for resistance group). For susceptibility, performance was high ($F1 = 0.993$).

Although models performed well with TF-IDF, transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) could be explored. Transformer-based models create contextualised text embeddings that capture semantic relationships and have been shown to outperform traditional methods (Alsentzer et al., 2019).

In a clinical context, the balance between precision and recall is critical. False positives may lead to inappropriate antibiotic use, contributing to AMR, while false negatives risk missing resistant infections and delaying effective treatment. Therefore, the high performance of the SVM model suggests potential for clinical application, although further validation on real-world data is required. This

could support the development of an NLP model that inputs extracted variables into existing clinical databases such as OMOP where issues of missing or incomplete data have been identified (White et al., 2019). This could enable healthcare data to be shared across institutions and support large-scale AMR surveillance.

6 Limitations

This study represents a controlled evaluation of how NLP models could be applied in organisations such as the NHS. It demonstrates how structured information can be extracted, anonymised and shared to support AMR prevention. However, evaluation on real clinical text is required with ethical approval.

The synthetic data was generated from a real, de-identified clinical dataset (ARMD), however this may not represent natural language and the variability and complexity of patient reports. This may contribute to the high performance observed. Outputs from LLMs are sensitive to prompts where the researcher unintentionally shapes the data based on assumptions and priorities, which introduces bias into models.

Only three resistance groups were explored which limits generalisation, and the balanced dataset does not reflect real-world class distributions where numbers of resistant strains are much lower. The datasets used were also relatively small, models should be evaluated on a larger scale to assess their scalability. Model interpretation is also important for clinical use, the outputs need to be understandable and justified.

7 Conclusion

This study demonstrates that NLP approaches can support the automated extraction of clinically meaningful AMR information from unstructured microbiology reports. Text preprocessing improved performance, and the SVM model achieved the best overall performance.

The results show that synthetic text, generated from a real de-identified clinical dataset, can be used to develop and evaluate NLP methods. However, the structured nature of synthetic data may overestimate performance compared to real-world clinical text.

Further evaluation and generalisation on real clinical data is required to support potential clinical application.

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